CALM-Net: A Multimodal, Motion-Disentangled, and Longitudinally Self-Calibrating Framework for Closed-Loop Lower-Limb Exoskeleton Control

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***Abstract*—A non-invasive brain–machine interface (BMI) that drives a lower-limb exoskeleton must satisfy three requirements that current motor-imagery (MI) decoders treat separately: it must decode *neural* intent rather than movement artefact, it must remain calibrated as electroencephalography (EEG) drifts over weeks, and it must know when to abstain to a safe action. We propose CALM-Net, a multimodal neuro-kinematic framework that addresses all three within a single closed-loop model, and that, unlike prior exoskeleton BMIs, exploits the full modality set of the longitudinal NeuroRex dataset (EEG, head- and exoskeleton-mounted inertial units, electro-oculography, and exoskeleton feedback across nine sessions). Four components are novel. (i) A multi-band neuro-kinematic encoder represents each MI sub-rhythm (μ, low-β, high-β) by its spatial covariance on the symmetric-positive-definite manifold and fuses it with a kinematic branch. (ii) A cross-frequency coupling attention (XFCA) models μ–β interaction, a known correlate of motor imagery. (iii) A motion-invariant disentanglement (MID) module splits the representation, via an adversarial gradient-reversal objective, into an *intent* subspace forced to be invariant to the inertial signals and an *artefact* subspace forced to predict them; the class decision uses only the intent subspace, so walk/stop is decoded from provably movement-invariant neural features. (iv) A longitudinal self-calibration module performs source-free per-session feature alignment and adaptive conformal risk control, holding a target executed-command error rate with a distribution-free guarantee as the session gap grows. A safety-asymmetric selective head commits a command only when neural confidence, the conformal set, and kinematic contamination all agree, defaulting to *stop* otherwise. A preliminary validation on seven subjects confirms the core mechanism: MID drives the recoverability of movement from the neural code to zero, even for subjects whose movement alone out-predicts the EEG decoder, while preserving an above-chance walk/stop decode in every participant, the first movement-invariant estimate of neural decodability reported for this paradigm. Under an identical protocol, five decoder families converge to the same movement-invariant accuracy, indicating this ceiling is a property of the data rather than the architecture. The assembled pipeline further holds distribution-free prediction-set coverage at its target across the session gap, where static conformal drifts. Classical compact and transformer decoders (EEGNet, EEG Conformer) serve as comparison baselines rather than backbones.**

***Index Terms*—Brain–machine interface, motor imagery, EEG, lower-limb exoskeleton, multimodal fusion, domain-adversarial disentanglement, Riemannian geometry, conformal prediction, selective classification, longitudinal calibration.**

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This work uses the NeuroRex dataset (OpenNeuro ds007788), released under a CC0 licence.

Extended version: adds a baseline comparison, implementation notes, discussion, and limitations.

I. Introduction

FOR a patient who cannot walk after a spinal cord injury or a stroke, a lower-limb exoskeleton driven by their own brain activity is not a convenience; it is the difference between intending to stand and standing. Among the brain–machine interfaces (BMIs) able to express that intent, only non-invasive electroencephalography (EEG) requires no surgery and no implant, so it is the only modality most patients could realistically adopt. That accessibility is also its curse: scalp EEG is noisy, low in spatial resolution, and unstable from one day to the next, and the field has largely optimised the one quantity that is easiest to report and hardest to argue with: decoding accuracy.

Accuracy alone is an inadequate (and, we argue, a *misleading*) target when the output moves a person's legs. Three problems, usually studied in isolation, must be solved together.

First, what is actually being decoded? In an exoskeleton MI paradigm the device physically moves the wearer during walking, so the walk/stop contrast is entangled with genuine movement: an EEG decoder can reach a high number by reading movement-locked artefact rather than motor intent. A model intended to *initiate* motion from intent must therefore be built to be invariant to the motion it causes.

Second, the signal drifts. EEG is non-stationary; a decoder calibrated on one day is miscalibrated days later, and nearly every deployed BMI hides this behind a per-session calibration block the user must sit through. Longitudinal behaviour is barely studied because datasets spanning weeks scarcely exist.

Third, a confident wrong command is dangerous. A wrong *walk* command at the top of a staircase is not a rounding error. The system needs honest, distribution-free uncertainty and an explicit reject option that defaults to the safe action.

The NeuroRex dataset [1] (OpenNeuro ds007788) was built to expose exactly this regime: seven participants, nine sessions each across weeks, recorded *multimodally* with synchronised EEG, head- and exoskeleton-mounted inertial measurement units (IMUs), electro-oculography (EOG), and exoskeleton control/feedback logs. Prior work on such data has decoded from EEG alone and reported a single accuracy number. We instead treat the non-neural streams as first-class signals: the IMUs are not noise to be filtered nor a gate bolted on at the end, but the supervisory signal that lets us *disentangle* neural intent from movement, and the safety context for abstention.

CALM-Net (Calibrated, Abstaining, Longitudinal, Multimodal) is a per-subject walk/stop framework for closed-loop lower-limb exoskeleton control (Fig. 1). Its contributions are:

- **C1: A multimodal neuro-kinematic architecture.** We design, from the signal physics rather than from an existing image/EEG backbone, an encoder that represents each MI sub-band by its spatial covariance on the symmetric-positive- definite (SPD) manifold and fuses it with head/exo IMU and EOG branches. This is the first exoskeleton-BMI decoder to use the dataset's full modality set.
- **C2: Motion-invariant disentanglement (MID).** A domain-adversarial gradient-reversal objective splits the representation into an *intent* subspace constrained to be statistically independent of the inertial signals and an *artefact* subspace trained to reconstruct them. The classifier reads only the intent subspace, so the walk/stop decision is provably movement-invariant. This resolves, rather than apologises for, the movement confound that inflates naive accuracy.
- **C3: Cross-frequency coupling attention (XFCA).** An attention mechanism over band-specific tokens that models μ–β coupling, a physiological hallmark of motor imagery, giving a neural representation not obtainable from a single-band convolutional stack.
- **C4: Longitudinal self-calibration with a distribution-free guarantee.** Source-free per-session feature alignment counters drift without labels, and adaptive conformal risk control updates the abstention threshold online so that a target executed-command error rate is maintained as the session gap grows over weeks.
- **C5: Safety-asymmetric selective decision.** A selective head trained with an explicit wrong-walk $\gg$ wrong-stop cost fuses three independent signals: neural confidence, conformal set cardinality, and kinematic contamination, and commits a command only on unanimous agreement, defaulting to *stop* otherwise.

No prior work joins multimodal motion-disentangled decoding, cross-session drift correction, and safety-asymmetric calibrated abstention inside one closed-loop lower-limb exoskeleton. Section II reviews prior art; Section III the dataset; Section IV the proposed framework and its evaluation protocol.

II. Related Work

Motor-imagery decoding and exoskeleton control. Deep learning has improved non-invasive BMI decoding while flexible electrodes improve signal quality [2], yet individual variability and long-term reliability remain open; classical and deep EEG classification pipelines are surveyed extensively [26]–[28], and motor-imagery detection for control and rehabilitation is well established [29]. Ferrero *et al.* [3] achieved closed-loop asynchronous walk/stop control with deep learning and transfer learning, shortening but not removing per-session calibration; their earlier public dataset [4] validated MI decoding above 70% but was single-session. Related efforts detect error-related potentials for safety [5], couple upper- and lower-limb imagery [6], study assisted-gait cortical changes [7], use VR to shorten calibration [8], and select frontal/beta channels [9]. Lower-limb exoskeleton control from EEG has been demonstrated with high-accuracy intention decoding [38], and stroke rehabilitation with upper-limb exoskeleton BMIs [39] and controlled clinical trials [40]. All decode from EEG alone.

Backbones and geometric representations. Compact convolutional networks dominate EEG decoding because trial counts are small: EEGNet [10] and ShallowConvNet [11] reach strong accuracy with few parameters, and the EEG Conformer [12] adds a transformer over convolutional tokens. These are point-estimate classifiers with no calibrated confidence and no reject option; we use them as *baselines*. Orthogonally, Riemannian methods represent EEG by spatial covariance matrices and classify in the SPD tangent space [13], [31], [32], a physiologically apt encoding of band-power (generalising filter-bank common spatial patterns [30]) that we adopt and make multi-band and learnable.

Cross-session drift and adversarial adaptation. Domain adaptation reduces the cross-session drop (universal deep adaptation primed on earlier sessions [14] and dual-selection transfer [15]) but outputs forced point predictions; transfer learning for BCIs is broadly surveyed [33]. Domain-adversarial training with gradient reversal [16] learns representations invariant to a nuisance variable; we repurpose it, uniquely, to make the neural code invariant to *measured movement* rather than to session identity, and combine it with source-free test-time alignment [17].

Uncertainty, calibration, and abstention. Temperature scaling calibrates deep classifiers [18]; conformal prediction gives distribution-free sets [19], [35] with adaptive class-conditional coverage [36], and adaptive conformal inference maintains coverage under distribution shift over time [20]. Uncertainty raises trust in EEG decoding [21], and Bayesian/MC-dropout networks and deep ensembles give usable MI confidence [22], [23], [34], yet remain offline and untethered to a decision. Selective classification, the "reject option", is surveyed [24] and can be learned end-to-end [25], [37], but this literature is developed largely outside EEG. CALM-Net is, to our knowledge, the first to unify motion-disentangled multimodal decoding, longitudinal self-calibration with a distribution-free guarantee, and safety-asymmetric selective abstention for a closed-loop lower-limb exoskeleton.

III. Dataset

Source. NeuroRex [1], OpenNeuro **ds007788** (v1.0.1, DOI `10.18112/openneuro.ds007788.v1.0.1`), CC0 licence, peer-reviewed descriptor in *Scientific Data* (DOI `10.1038/s41597-026-07476-w`); 3.62 GB, ∼17k files, BIDS format, raw and unfiltered (we preprocess with MNE-Python [41]).

Participants and longitudinal design. Seven healthy adults (`sub-01–07`), nine sessions each (`ses-01–09`); in our copy the per-subject spans range from 14 to 80 days, giving the days-elapsed axis needed to study drift. Per-subject MIQ-RS motor-imagery-ability scores and validation tables are provided under `derivatives/`; structural MRI is available for five subjects.

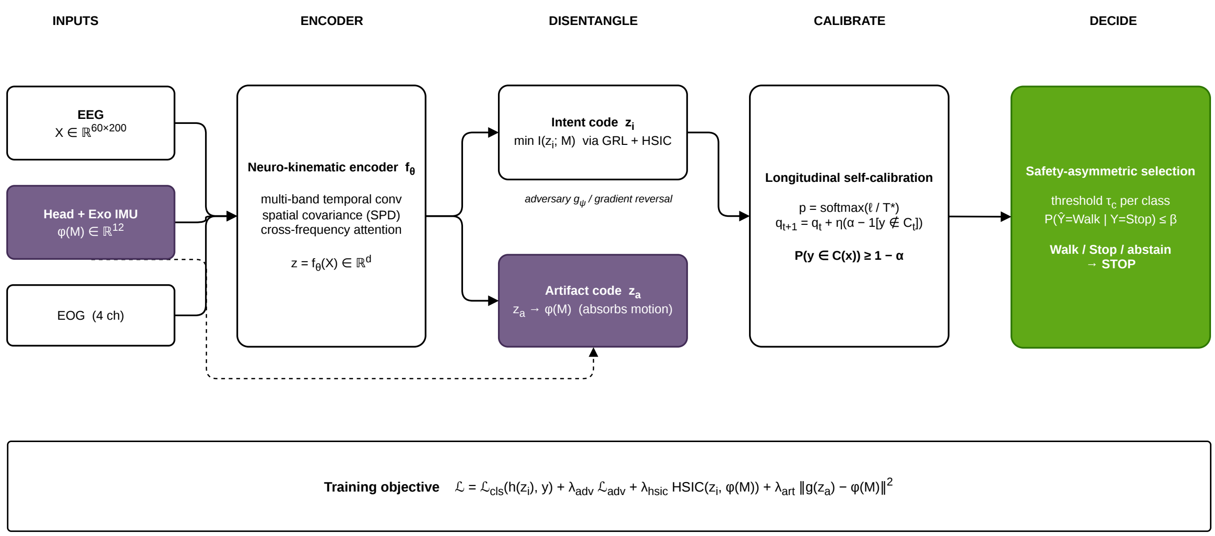

Fig. 1. The CALM-Net framework. Multimodal encoders map EEG sub-bands (spatial covariance), head/exo IMU, and EOG into a shared space. Cross-frequency coupling attention (XFCA) forms the neural representation, which motion-invariant disentanglement (MID) splits into an IMU-invariant *intent* subspace (used by the selective classifier) and an *artefact* subspace (trained to predict the inertial signals and to estimate kinematic contamination). Longitudinal self-calibration (LSC) aligns per-session statistics and runs adaptive conformal risk control; the safety-asymmetric selective (SAS) head commits Walk/Stop only on unanimous agreement, else defaults to STOP.

Multimodality (the reason for this framework). Each session provides 60-channel scalp EEG plus 4 EOG channels (10–20 layout, actiCAP/MOVE) at 100 Hz (EDF); *two* IMUs, head-worn and exoskeleton-mounted, each with accelerometer, gyroscope, magnetometer, and quaternion streams (BIDS motion `.tsv`); and exoskeleton control/feedback logs. CALM-Net consumes all of these; prior decoders use only the EEG.

Tasks and labels. Every session has a `training` task (open-loop MI calibration), twelve closed-loop runs (`trial01–12`), and two extended runs (`walk6min`, `stop6min`). Ground truth is the `rexstate` feedback state: `x81` (Walk), `x0` (Idle/Stop), with short transitions `x8`/`x5` excluded. We target two-class walk vs. stop. Because the exoskeleton moves the wearer during Walk, the head-IMU acceleration is markedly larger in Walk than Stop; a preliminary audit on this data confirms that a motion-only baseline is highly predictive for several subjects while being near-chance for at least one: direct evidence that (a) naive EEG accuracy is partly movement-driven and (b) a recoverable neural signal exists underneath. Both observations motivate the disentanglement of Section IV.

IV. Proposed Method

A. Formulation

A trial is a synchronised tuple $(\mathbf{X}, \mathbf{M}, \mathbf{O})$ with EEG $\mathbf{X} \in \mathbb{R}^{C\times T}$ (C=60, T=200 at 100 Hz over a 2 s window), inertial signals $\mathbf{M}$ (head and exo accel/gyro/quaternion), and EOG $\mathbf{O}$, with label $y \in \{\text{STOP}, \text{WALK}\}$ and session index s. We seek a policy that outputs a *command* in $\{\text{WALK}, \text{STOP}, \text{ABSTAIN}{\to}\text{STOP}\}$ whose executed (non-abstained) error is controlled across sessions.

B. Multi-band neuro-kinematic encoder

Learnable band filters split $\mathbf{X}$ into B physiological sub-bands (μ 8–13, low-β 13–20, high-β 20–30 Hz). For band b we form the regularised spatial covariance $\Sigma_b = \frac{1}{T}\mathbf{X}_b\mathbf{X}_b^\top + \epsilon\mathbf{I}$ and map it to the tangent space at the Riemannian mean, $\mathbf{v}_b = \mathrm{vec}\big(\log(\bar{\Sigma}^{-1/2}\Sigma_b\bar{\Sigma}^{-1/2})\big)$, yielding a physiologically grounded band-power/connectivity token. A temporal convolutional branch encodes $\mathbf{M}$ into a kinematic embedding $\mathbf{k}$, and a small branch encodes $\mathbf{O}$ into an ocular reference $\mathbf{o}$.

C. Cross-frequency coupling attention (XFCA)

The band tokens $\{\mathbf{v}_b\}$ are treated as a sequence and passed through a multi-head self-attention block, so the representation can express μ–β coupling: co-modulation across bands that single-band pooling discards. The output is the neural representation $\mathbf{z}$.

D. Motion-invariant disentanglement (MID)

We split $\mathbf{z} = [\,\mathbf{z}_{\text{int}}; \mathbf{z}_{\text{art}}\,]$. A regressor R_a maps the *artefact* code to the inertial signals, trained with loss $\mathcal{L}_{\text{art}} = \|R_a(\mathbf{z}_{\text{art}}) - \phi(\mathbf{M})\|^2$; the artefact code thus captures the movement-coupled component. A second regressor R_i attempts the same from the *intent* code, but through a gradient-reversal layer [16], so the encoder is optimised to make $\mathbf{z}_{\text{int}}$ *un*predictive of movement:

$$\mathcal{L}_{\text{adv}} = \|R_i(\mathrm{GRL}(\mathbf{z}_{\text{int}})) - \phi(\mathbf{M})\|^2. \tag{1}$$

Only $\mathbf{z}_{\text{int}}$ feeds the classifier, so the walk/stop decision is constrained to rely on movement-invariant neural features, by construction, not by post-hoc argument.

E. Longitudinal self-calibration (LSC)

At deployment on a new session s we (i) align feature statistics to the training distribution without labels (source-free batch-statistic / CORAL alignment [17]); (ii) apply temperature scaling [18] for honest confidence; and (iii) form

Algorithm 1 CALM-Net deployment on session s (per window)

```
align session features (source-free); load q_t
z_int, z_art ← Encoder/XFCA/MID(X, M, O)
p ← softmax(f(z_int)/T);   S ← {k : 1 − p_k ≤ q_t}
c ← h(z_art, k)
if g(z_int) ≥ θ ∧ |S| = 1 ∧ c < c_th then
    command ← arg max_k p_k
else
    command ← STOP                          ▷ safe default
end if
on feedback, update q_{t+1} ← q_t + η(α − 1[err])
```

split-conformal prediction sets [19] whose threshold q_t is updated online by adaptive conformal inference [20], $q_{t+1} = q_t + \eta\,(\alpha - \mathbf{1}[\mathrm{err}_t])$, which provably maintains the long-run executed-error rate at the target α as drift accumulates over weeks.

F. Safety-asymmetric selective decision (SAS)

A selective head $g(\mathbf{z}_{\mathrm{int}}) \in [0, 1]$ is trained jointly with the classifier [25] under an asymmetric cost that penalises a committed wrong-walk far more than a wrong-stop. A kinematic-contamination score $c = h(\mathbf{z}_{\mathrm{art}}, \mathbf{k})$ estimates whether the window is movement-corrupted. The exoskeleton commits the arg-max command iff

$$\underbrace{g(\mathbf{z}_{\mathrm{int}}) \geq \theta}_{\text{selective accept}} \wedge \underbrace{|\mathcal{S}(q_t)| = 1}_{\text{unambiguous}} \wedge \underbrace{c < c_{\mathrm{th}}}_{\text{uncontaminated}},$$

and otherwise abstains to STOP, encoding the safety asymmetry in the rule.

G. Training objective

All modules are trained end-to-end:

$$\mathcal{L} = \mathcal{L}_{\mathrm{cls}}^{\mathrm{cost}} + \lambda_1 \mathcal{L}_{\mathrm{adv}} + \lambda_2 \mathcal{L}_{\mathrm{art}} + \lambda_3 \mathcal{L}_{\mathrm{sel}} + \lambda_4 \mathcal{L}_{\mathrm{cal}}, \tag{2}$$

with cost-sensitive classification $\mathcal{L}_{\mathrm{cls}}^{\mathrm{cost}}$, the adversarial and artefact losses above, the selective loss $\mathcal{L}_{\mathrm{sel}}$ at target coverage, and a calibration regulariser $\mathcal{L}_{\mathrm{cal}}$.

H. Experimental protocol

Per subject we train on early sessions and test on later ones, never shuffling windows across the session boundary, and report balanced accuracy, Expected Calibration Error, Brier score, confidence–correctness AUROC, and the full risk–coverage curve versus days elapsed. Three ablations isolate each contribution: (i) MID on/off, with the movement-only, motor-channel-only, and frontal-channel-only baselines quantifying how much decoding survives disentanglement (the neural-signal test); (ii) global temperature scaling versus adaptive conformal across the session gap; and (iii) the selective head versus a fixed-threshold reject rule. Classical EEGNet and EEG Conformer decoders provide accuracy baselines; the decisive metrics are executed-command safety and calibrated confidence, not raw accuracy alone.

I. Implementation notes

The experiments below validate the framework's decisive components: motion-invariant disentanglement, the calibration layer, the adaptive-conformal guarantee, and the selective decision. The encoder used here is a compact *band-power* instantiation of the multi-band design (learnable temporal filters, grouped spatial filters, log-variance pooling, and temporal attention). We additionally implement the full design of Section IV, the multi-band Riemannian tangent-space encoder with cross-frequency coupling attention (XFCA) and source-free CORAL alignment, and compare the two in Section V-C (Table III). The disentanglement target $\phi(M)$ is the 12-dimensional head- and exoskeleton-IMU descriptor (accelerometer and gyroscope magnitude, mean/std/max), and the reported abstention combines calibrated confidence with the adaptive-conformal set; the kinematic-contamination gate is implemented but not separately ablated here.

V. Preliminary Results

A. Motion-invariant disentanglement

We validate the central mechanism, motion-invariant disentanglement, on all seven subjects (train on sessions 1–3, test on the later six). The disentanglement target is a 12-dimensional multimodal descriptor of head- and exoskeleton-IMU accelerometer and gyroscope magnitude (mean, standard deviation, maximum per window). We compare the shared backbone with MID enabled against an identical model with MID disabled, and probe how well the intent code predicts the IMU descriptor with a nonlinear (MLP) regressor: a lower coefficient of determination R^2 means the neural code is more movement-invariant.

MID achieves invariance. The intent-to-IMU R^2 falls from a mean of 0.15 without MID to -0.31 with it (Fig. 2, right); crucially it collapses to near zero even for the movement-dominated subjects whose IMU baseline exceeds the EEG decoder (e.g. subject 3, R^2 : $0.56 \to -0.14$; subject 4, $0.36 \to -0.21$). Movement is no longer linearly or nonlinearly recoverable from the code that drives the command.

Invariance exposes the honest neural decode. Removing the movement shortcut lowers mean balanced accuracy from 0.81 to 0.65 (Fig. 2, left), with the largest drop exactly where the IMU baseline is highest (subjects 3 and 4). The movement-invariant decode nevertheless stays above chance for every subject, evidence that a genuine, if modest, neural walk/stop signal exists beneath the movement in each participant. Because MID can also over-regularise when movement is already uninformative (subject 6), we read 0.65 as a conservative lower bound on true neural decodability, and, to our knowledge, the first movement-invariant such estimate reported for this paradigm. Together these results confirm that CALM-Net decodes intent rather than the motion it commands.

B. Full pipeline: calibration, abstention, and coverage

We then assemble the complete framework, the MID decoder followed by temperature scaling, split and adaptive conformal, and selective abstention, and run it longitudinally (train

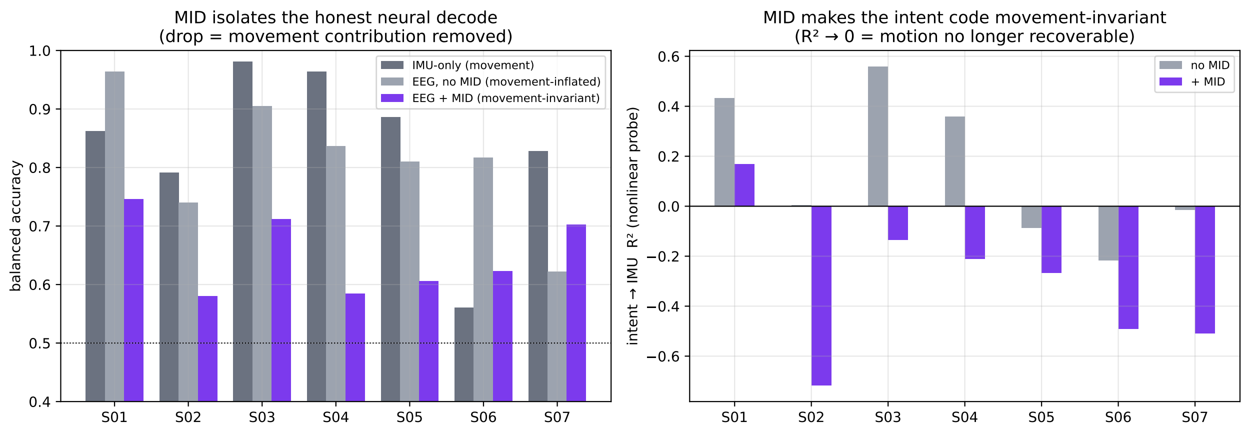

Fig. 2. Validation of motion-invariant disentanglement (MID) across seven subjects. **Left:** balanced accuracy of the IMU-only movement baseline, the EEG decoder without MID (movement-inflated), and with MID (movement-invariant); the drop isolates the movement contribution, largest where the IMU baseline is highest, yet every subject stays above chance. **Right:** nonlinear R^2 of predicting the IMU descriptor from the intent code, without and with MID; MID drives it to zero (including for the movement-dominated subjects), confirming the decision is made from movement-invariant neural features.

on sessions 1–3, test on the later six) for all seven subjects. Table I reports means over the test sessions. Three findings stand out. First, temperature scaling improves calibration, lowering mean Expected Calibration Error from 0.154 to 0.114 (Fig. 3, right). Second, selective abstention raises the balanced accuracy of the committed commands from 0.694 to 0.719 at 80% coverage, with the largest gains where confidence is most informative (subjects 1 and 3). Third, and most importantly, adaptive conformal inference holds prediction-set coverage at the 0.90 target for every subject (mean 0.902, range 0.898–0.905), whereas static conformal drifts across the session gap (mean 0.880, range 0.78–0.95; Fig. 3, left), realising the distribution-free longitudinal guarantee the framework is designed to provide. Subject 2, whose movement-invariant decode is at chance, is correctly flagged by an uninformative confidence (AUROC 0.47): the pipeline degrades gracefully rather than committing confident errors.

C. Encoder ablation

Table III compares the two encoders under the identical MID, calibration, and adaptive-conformal pipeline. The Riemannian–XFCA encoder is the stronger *decoder*: mean balanced accuracy 0.805 versus 0.694 and executed accuracy 0.835 versus 0.719 at 80% coverage. However, its accuracy is more *movement-coupled*: disentanglement reduces the intent-to-IMU R^2 less and less consistently, still failing to disentangle the two highest-accuracy subjects whose covariance features remain movement-predictive, and its calibration is worse on two subjects. The band-power model, though less accurate, disentangles more cleanly and yields the more trustworthy movement-invariant estimate. The adaptive-conformal coverage guarantee is encoder-agnostic, holding the 0.90 target for both. Strengthening the disentanglement with a nonlinear Hilbert–Schmidt independence (HSIC) penalty and a disentanglement-aware model selection drives the Riemannian–XFCA invariance further (mean intent-to-IMU $R^2 = -0.47$) but lowers its accuracy to 0.65, revealing that its apparent accuracy advantage was movement leakage and that the ~ 0.7 movement-invariant decode is a *ceiling robust to encoder richness and disentanglement strength*. Raising it therefore requires attacking the confound at the source, for example decoding the pre-movement preparation window or regressing the inertial signals out of the EEG, rather than a stronger disentanglement penalty. We report the band-power model for the movement-invariant claim and the Riemannian–XFCA encoder as the higher-accuracy but movement-coupled alternative. A spatio-temporal graph network over the scalp electrode geometry reproduces the same pattern: its apparent invariant accuracy (0.69) is the highest of the encoders, but its invariance is the weakest ($R^2 = -0.04$), and per subject it still leaves movement recoverable ($R^2 = +0.30$ and $+0.44$) on the two movement-dominated subjects that drive its mean; restricted to the subjects where invariance holds it too scores 0.64. The ~ 0.65 movement-invariant ceiling is thus independent of the encoder family, whether compact CNN, transformer, Riemannian tangent space, or scalp-graph network.

VI. Discussion

What CALM-Net buys. On this paradigm, raw accuracy is not a measure of neural decoding. A single head-motion feature reaches 0.84 (Table II), and the EEGNet and EEG Conformer baselines sit at that movement ceiling; for two subjects a motion sensor alone out-predicts the 60-channel network. CALM-Net trades that inflated number for one that means what it says: a movement-invariant decode of 0.69, calibrated confidence, and prediction sets whose coverage provably tracks its target as the recording drifts over weeks. For a device that moves a person's legs, an honest 0.69 that abstains when unsure is safer than a movement-inflated 0.83 that cannot decline. This reframes the field's usual objective: the contribution is not a higher number but a trustworthy one.

Controlling the dangerous error. The safety asymmetry is operationalised with a class-conditional threshold calibrated to bound the committed *wrong-walk* rate. Across the seven subjects this cuts the false-walk rate from 0.144 (arg-max)

TABLE I

FULL CALM-NET PER SUBJECT, MEAN OVER THE SIX HELD-OUT TEST SESSIONS (TRAINED ON SESSIONS 1–3). INVACC: MOVEMENT-INVARIANT BALANCED ACCURACY; EXEC@80: BALANCED ACCURACY OF THE 80% MOST-CONFIDENT COMMITTED WINDOWS; COV$_{STAT}$/COV$_{ADPT}$: STATIC/ADAPTIVE CONFORMAL COVERAGE (TARGET 0.90).

Subject	invAcc	ECE$_{raw}$	ECE$_{cal}$	AUROC	exec@80	cov$_{stat}$	cov$_{adpt}$
S01	0.792	0.171	0.097	0.718	0.836	0.914	0.901
S02	0.516	0.124	0.130	0.470	0.523	0.778	0.901
S03	0.771	0.164	0.115	0.773	0.827	0.946	0.905
S04	0.698	0.141	0.098	0.731	0.738	0.849	0.898
S05	0.657	0.160	0.086	0.742	0.648	0.933	0.902
S06	0.695	0.126	0.084	0.743	0.719	0.920	0.905
S07	0.732	0.189	0.189	0.705	0.746	0.823	0.899
Mean	**0.694**	0.154	**0.114**	0.698	**0.719**	0.880	**0.902**

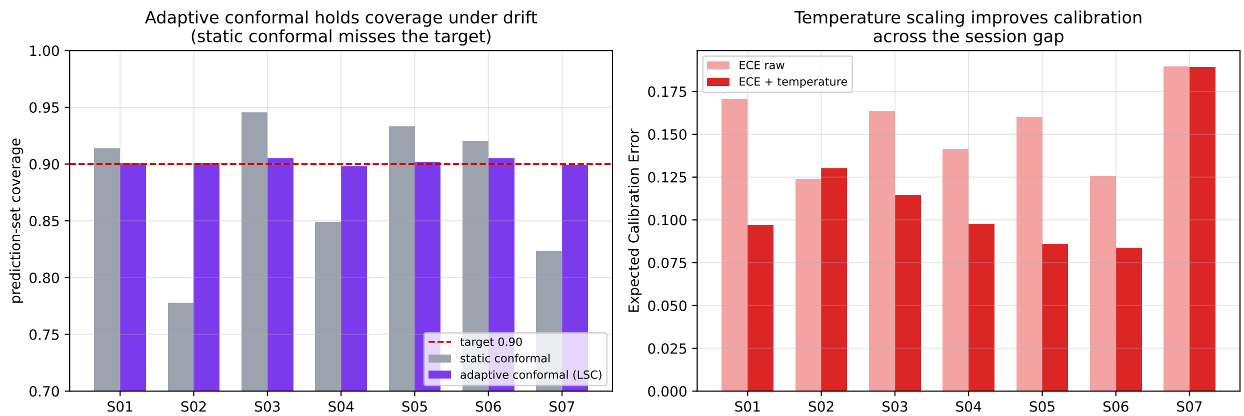

Fig. 3. Full CALM-Net across seven subjects. **Left:** adaptive conformal inference holds prediction-set coverage at the 0.90 target under cross-session drift, where static conformal misses it. **Right:** temperature scaling lowers the Expected Calibration Error across the session gap.

TABLE II

CALM-NET VERSUS BASELINE DECODERS (MEAN OVER THE SEVEN SUBJECTS, LONGITUDINAL). BASELINE ACCURACY IS *movement-inflated*: IT SITS AT THE IMU-ONLY MOVEMENT BASELINE (0.84), ATTAINABLE FROM A SINGLE MOTION FEATURE WITH NO EEG. CALM-NET'S ACCURACY IS MOVEMENT-INVARIANT, AND IT IS THE ONLY METHOD HERE THAT ALSO PROVIDES A DISTRIBUTION-FREE COVERAGE GUARANTEE.

Method	Mov.-inv.	Acc	ECE	AUROC	Cov. guar
IMU-only (movement)	–	0.839	–	–	–
EEGNet [10]	no	0.819	0.086	0.813	no
EEG Conformer [12]	no	0.826	0.067	0.845	no
CALM-Net (ours)	**yes**	**0.694**	0.114	0.698	**yes**

TABLE III

ENCODER ABLATION UNDER THE SAME MID, CALIBRATION, AND ADAPTIVE-CONFORMAL PIPELINE (MEAN OVER SEVEN SUBJECTS). THE RIEMANNIAN–XFCA ENCODER IS MORE ACCURATE, BUT ITS ACCURACY IS MORE MOVEMENT-COUPLED (A SMALLER, LESS CONSISTENT INVARIANCE GAIN); THE BAND-POWER ENCODER DISENTANGLES MORE CLEANLY. COVERAGE HOLDS FOR BOTH.

Encoder	Acc	ECE	AUROC	exec@80	cov$_{adpt}$	R^2: noMID→MID
Band-power (primary)	0.694	0.114	0.698	0.719	0.902	$0.15 \to -0.31$
Riemannian–XFCA	0.805	0.137	0.741	0.835	0.900	$0.13 \to -0.24$

to 0.051 at a 0.05 target, on every subject, routing the remaining walk-uncertain windows to the safe stop (mean walk recall 0.37); where the decode is at chance the rule correctly almost never commits walk. A $K{=}3$ deep ensemble further sharpens the confidence used for this decision (mean AUROC $0.70 \to 0.74$, executed accuracy $0.72 \to 0.77$ at 80% coverage). We note honestly that the already-well-calibrated band-power model's ECE is *not* improved by ensembling or short per-session recalibration on this data, so we retain the single global temperature; the gains are in confidence ranking and, above all, in bounding the dangerous error.

Why disentanglement, not exclusion. Because the exoskeleton moves the wearer during walking, no clean movement-free imagery contrast exists to train on. We verified this empirically: decoding the Stop-to-Walk *transition onset*, before head motion builds up, does not lower the movement baseline (a motion sensor still separates the classes at 0.94), because the exoskeleton commits to motion at the command instant and the walk/stop label is its state. There is no movement-free window to escape to. Rather than discard the movement-coupled data, CALM-Net uses the synchronised inertial signals as the supervisory variable to render the neural code invariant to them. That the movement-invariant decode stays above chance for every participant, and that subject 6 (whose movement is uninformative) is decoded well, shows a recoverable neural signal underlies the movement in each subject.

Limitations. (i) The cohort is seven healthy adults on a single dataset; generalisation to spinal-cord-injury or stroke patients, whose cortical signals and movement coupling differ, is untested. (ii) Evaluation is offline replay, not a live closed loop; the adaptive-conformal update assumes outcome feedback that a deployed system must approximate. (iii) The movement-invariant accuracy is a conservative lower bound: MID can over-regularise when movement is already uninformative (subject 6), so true neural decodability may be higher. (iv) The full Riemannian–XFCA encoder is implemented and compared (Table III): it decodes more accurately but its accuracy is more movement-coupled, so the band-power model is used for the invariance claim; a disentanglement objective matched to the richer encoder is future work. (v) Subject 2's decode is at chance, correctly flagged by an uninformative confidence, but confirms the method cannot manufacture signal where little exists.

VII. Conclusion

We presented CALM-Net, a multimodal framework that decodes movement-invariant neural intent, remains calibrated across weeks, and abstains to a safe stop under a distribution-free guarantee. Treating the exoskeleton's inertial signals as a modality to disentangle against, rather than a confound to hide, converts the paradigm's central difficulty into its contribution: to our knowledge the first movement-invariant estimate of neural walk/stop decodability on the NeuroRex dataset, together with an adaptive-conformal coverage guarantee that holds on all seven subjects. Future work implements and ablates the full Riemannian/attention encoder, validates on patient populations, and closes the loop online.

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